

Titanic Dataset Analysis Report

Datasets Used : train.csv, test.csv and gender_submission.csv

1. Objective

The goal of this analysis is to explore the Titanic dataset to identify trends, patterns, and relationships that influence passenger survival. We aim to understand how features like gender, class, age, and fare impacted the survival rate.

2. Dataset Overview

The dataset is split into training and test sets.

- The training dataset contains 891 rows and 12 columns.
- The test dataset, after merging with the gender submission file, contains 418 rows.

Key features include PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked.

3. Data Exploration

Key findings from the `.info()` and `.describe()` methods:

- Age and Cabin columns have missing values.
- Fare values vary significantly, indicating class differences.
- Pclass and Sex show clear patterns in relation to survival.

Top value counts:

- Survived: 549 did not survive, 342 survived.
- Sex: 577 males, 314 females.
- Embarked: S (644), C (168), Q (77)

4. Visualizations & Observations

Correlation Heatmap:

- Positive correlation between Fare and Survived.
- Negative correlation between Pclass and Survived.
- Weak correlation with other features.

Pairplot:

- Survivors are more likely to be in Pclass 1, younger, and with higher Fare.

Histograms and Boxplots:

- Age distribution shows more survivors in the 20–40 age range.
- Higher Fare passengers had higher survival rates.
- Pclass 1 had both older and younger passengers.
- Passengers paying higher fares had better survival chances.

5. Key Findings

1. Gender:

- Females had a significantly higher survival rate than males.

2. Passenger Class:

- Passengers in 1st class had better survival outcomes.

3. Age:

- Children and younger passengers had a better chance of survival.

4. Fare:

- Those who paid higher fares generally survived more.

5. Embarked Location:

- Embarking from Cherbourg (C) was linked to higher survival.

6. Family:

- Having family aboard (SibSp or Parch > 0) showed slight survival advantages.

6. Conclusion

Survival on the Titanic was influenced by multiple factors. Females, children, and first-class passengers had better chances of survival. Fare and embarked location also played significant roles. These insights can guide predictive modeling efforts for Titanic survival prediction.